

PHASE 4 – PROJECT PLANNING PHASE

EduGenie: Google Gemini Powered Learning Assistant

Team ID	SWTID-2026-1665
Team Size	5
Team Leader	Akalya
Team Members	Naveena, Reshma, Sathya, Sudharsana K

1. Planning Logic

Sprint: A fixed period in which the team works to complete a specific set of tasks.

Epic: A large project feature divided into smaller User Stories.

User Story: A feature described from the user's point of view.

Story Point: Represents the effort required to complete a User Story.

Story Point	Difficulty
1	Very Easy
2	Easy
3	Moderate
5	Difficult
8	Very Difficult

2. Project Milestones & Tasks

Epic	Milestone
Epic 1	User Authentication & Profile
Epic 2	AI Learning Assistant
Epic 3	Learning & Assessment
Epic 4	Progress Tracking & Dashboard
Epic 5	Testing, Integration & Deployment

Major Tasks:

Epic 1 – User Authentication & Profile: User registration, user login, student profile creation, profile management.

Epic 2 – AI Learning Assistant: Enter learning question, Gemini-powered response, simple explanations, subject/topic-based learning, conversation history.

Epic 3 – Learning & Assessment: Generate quiz questions, answer questions, display quiz results, learning resources/recommendations.

Epic 4 – Progress Tracking & Dashboard: Track completed topics, display quiz scores, learning progress, student dashboard.

Epic 5 – Testing & Deployment: Functional testing, AI response testing, UI testing, integration testing, bug fixing, final deployment.

3. Product Backlog

Sprint	Epic	USN	User Story / Task	Points	Priority	Team Member
Sprint 1	User Authentication	USN-1	As a student, I can register an account	2	High	Akalya
Sprint 1	User Authentication	USN-2	As a student, I can securely login	2	High	Naveena
Sprint 1	User Profile	USN-3	As a student, I can create and update my profile	2	Medium	Reshma
Sprint 1	AI Assistant	USN-4	As a student, I can enter a question to the AI assistant	3	High	Sathya
Sprint 2	AI Assistant	USN-5	As a student, I can receive AI-generated explanations	5	High	Sudharsana
Sprint 2	AI Assistant	USN-6	As a student, I can select a subject/topic for learning	3	High	Naveena
Sprint 2	AI Assistant	USN-7	As a student, I can view previous conversations	3	Medium	Reshma
Sprint 3	Assessment	USN-8	As a student, I can generate quizzes on a topic	5	High	Akalya

Sprint 3	Assessment	USN-9	As a student, I can submit quiz answers	2	High	Sathya
Sprint 3	Assessment	USN-10	As a student, I can view my quiz score	2	High	Sudharsana
Sprint 3	Resources	USN-11	As a student, I can receive learning resources	3	Medium	Naveena
Sprint 4	Dashboard	USN-12	As a student, I can view my learning progress	5	High	Reshma
Sprint 4	Dashboard	USN-13	As a student, I can view my scores and completed topics	3	High	Akalya
Sprint 4	Testing	USN-14	As a user, I can use the application without major errors	5	High	All Members
Sprint 4	Deployment	USN-15	As a team, we can deploy the completed application	3	High	All Members

4. Sprint Delivery Plan

Sprint 1 – User Setup & Basic AI Interface

Goal: Create basic application structure and student authentication.

User Story	Points
USN-1 – Registration	2
USN-2 – Login	2
USN-3 – Student Profile	2
USN-4 – Enter AI Question	3
Total	9

Sprint Deliverable: Basic EduGenie application with registration, login, profile and initial AI interaction interface.

Sprint 2 – AI Learning Assistant

Goal: Develop the core Gemini-powered learning functionality.

User Story	Points
USN-5 – AI-generated explanations	5
USN-6 – Subject/Topic Selection	3
USN-7 – Conversation History	3
Total	11

Sprint Deliverable: Functional AI learning assistant for topic-based explanations.

Sprint 3 – Quiz & Learning Resources

Goal: Add assessment and learning-support features.

User Story	Points
USN-8 – Generate Quiz	5
USN-9 – Submit Answers	2
USN-10 – Display Score	2
USN-11 – Learning Resources	3
Total	12

Sprint Deliverable: Students can generate quizzes, answer them, view scores and access resources.

Sprint 4 – Dashboard, Testing & Deployment

Goal: Complete progress tracking, testing and deployment.

User Story	Points
USN-12 – Learning Progress	5
USN-13 – Scores & Completed Topics	3
USN-14 – Testing & Bug Fixing	5
USN-15 – Deployment	3
Total	16

Sprint Deliverable: Complete application with dashboard, progress tracking, testing and deployment.

5. Sprint Summary

Sprint	Main Focus	Story Points
Sprint 1	Authentication & Basic Interface	9
Sprint 2	Gemini AI Learning Assistant	11
Sprint 3	Quiz & Learning Resources	12
Sprint 4	Dashboard, Testing & Deployment	16
Total	Complete Project	48

6. Project Progress Tracking

Task	Status	Progress
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Registration	Done	100%
Login	Done	100%
Student Profile	Done	100%
AI Question Interface	In Progress	70%
Gemini Response	In Progress	60%
Quiz Generation	To Do	0%
Progress Dashboard	To Do	0%
Testing	To Do	0%
Deployment	To Do	0%

Note: The percentages above are sample project-tracking values and should be updated with the team's actual progress.

7. Team Management Tool – Trello

For Agile project management, the team can use Trello.

Board List	Example Cards
TO DO	Quiz Generation; Progress Dashboard; Learning Resources
IN PROGRESS	Gemini AI Integration; Conversation History
TESTING	Login Testing; AI Response Testing
DONE	Registration; Student Profile; Basic UI

Each task can be assigned to a team member with a deadline and priority.

8. Team Responsibility

Member	Planned Responsibility
Akalya – Team Leader	Project coordination, authentication, dashboard
Naveena	Subject selection, resources, AI module support
Reshma	Profile, conversation history, progress tracking
Sathya	AI interface, quiz submission, UI support
Sudharsana	Gemini AI integration, AI responses, quiz scoring & testing

9. Milestone Schedule

Sprint 1 → User Registration → Login → Profile → Basic AI Interface

Sprint 2 → Gemini Integration → AI Explanation → Subject Selection → History

Sprint 3 → Quiz Generation → Quiz Answer → Score → Learning Resources

Sprint 4 → Progress Dashboard → Testing → Bug Fixing → Deployment

Final Project Outcome

EduGenie will provide students with a Google Gemini-powered learning assistant that supports question answering, concept explanation, topic-based learning, quizzes, learning resources and progress tracking.